Welcome to NRES 470/670

Applied Population Ecology, Spring 2026

Instructor: Kevin Shoemaker
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Teaching Assistant Jerrod Merrell (jmerrell@unr.edu)
TA Office hours: TBD

Course Meeting Times

Lecture & Discussion: M, W at 10am (50 mins) in FA 301 (Fleischmann Ag Building)
Lab: F at 1pm (2 hrs 45 mins) in FA 301

Course description

This class explores how concepts of population ecology can be used to inform the conservation and management of natural populations and ecosystems. We emphasize practical approaches to problem-solving in ecology, conservation, and wildlife management using simulation models and statistics. Topics include Population Viability Analysis (PVA), limits to population growth, metapopulation ecology, species interactions (competition and predation), threats to wild populations, wildlife management and more. Laboratory exercises provide students with hands-on experience with wildlife population models and their practical applications in wildlife ecology and management.

Prerequisites

- BIOL 314 or NRES 217 (Ecology)
- NRES 310 (Wildlife Ecology and Management)
  NOTE: this course is a prerequisite for NRES 488 (Dynamics and Management of Wildlife Populations) and is designed to complement NRES 421 (Conservation Biology).

Texts

- Gotelli, N. J. A primer of ecology

- Additional readings will be assigned for discussion periodically.

Software

- InsightMaker- web-based systems modeling tool(free, web-based, need account)

- R- software for statistical computing and graphics (free installation)
- Spreadsheet software (MS Excel, Google Sheets or equivalent)
- Top Hat (interactive classroom software- invitations should have been emailed to you)

Student Learning outcomes

SLO 1. Explain how and why simulation models are used by ecologists and wildlife professionals.
SLO 2. Apply tools such as population viability analysis (PVA) and metapopulation models to address the conservation and management of wild populations.
SLO 3. Perform basic statistics, data visualization, simulation modeling and model validation with Excel, the statistical computing language 'R', and the web-based software, InsightMaker.
SLO 4. Critically evaluate the strength of inferences drawn from ecological simulation models using tools such as sensitivity analysis.
SLO 5. Explain how species interactions can influence population dynamics (e.g., predictions of species range shifts).
SLO 6. Communicate original research in applied population and community ecology via professional-style oral and written presentations.

Grading:

The course grade will be based on the following components:

- Lab exercises (7 total) 20%
  Lectures/participation 10%
  Group project 30%
  Midterm exam # 1 (date TBD) 10%
  Midterm exam # 2 (date TBD) 10%
  Final exam 20%

Grading scale: A (100 to 93), A- (92 to 90), B+ (89 to 87), B (86 to 83), B- (82 to 80), C+ (79 to 77), C (76 to 73), C- (72 to 70), D+ (69 to 67), D (66 to 63), D- (62 to 60), F (below 60).

Exams:

There will be two midterm exams and a final exam, all of which will be cumulative, covering all course material covered up to the week prior to the exam. These will consist of multiple-choice, short-answer questions, and essay questions requiring synthesis of key concepts.

Lectures

Lecture grades will be based primarily on participation and short in-class quizzes (via TopHat). Participation is essential to the learning process (and to our mutual enjoyment of this class). Learning is not a passive process; students are expected to engage with the material in class rather than simply listen and take notes. You should be prepared in class to ask questions, to answer questions, and to engage in problem-solving activities.

Labs

Lab exercises will focus on applying concepts and methods introduced in lectures, and will involve real-world problems in wildlife conservation and management wherever possible. Graded lab assignments will involve figures, tables, InsightMaker models and R code (when applicable) and responses to questions in short-answer format.

Final group project

Students will work in groups of 3-4 to perform a population viability analysis (PVA) to rank conservation or management actions for a species of conservation concern (species of your choice!). Grading will be based on finished products (written and oral presentations) as well as participation and peer evaluations.

Graduate credit (for students enrolled in NRES 670)

Graduate students will be subject to additional expectations in order to receive graduate credit for this course. In particular, graduate students will be expected to develop an original lecture and lead an original lab activity related to a topic relevant to wildlife population ecology. Graduate students will also be expected to achieve a deeper understanding of the course material, and therefore will be expected to participate as leaders in discussions and lab activities.

Make-up policy and late work:

Missed exams and labs cannot be made up, except in the case of emergencies. If you miss a class meeting, it is your responsibility to talk to one of your classmates about what you missed. If you miss a lab meeting, you are still responsible for completing the lab activities and write-up. Let me or your TA know in advance if you are going to miss class or lab.

Top Hat

We will be using the Top Hat interactive learning platform in class. You will be able to submit answers to in-class questions using Apple or Android smartphones and tablets, laptops, or through text message.

Top Hat is free of charge for UNR students this year! You should all have received an invitation via email. You can also enroll using the join code 866526.

Course Schedule

NOTE: schedule is subject to change- please check the course website for the latest updates!

Week	Dates	Topic	Readings
Week 1	1/22/2024	LECTURE: Course overview; Intro to Systems Thinking	
	1/24/2024	LECTURE: Intro to Population Ecology; Exponential growth	Gotelli Chapter 1

Week	Dates	Topic	Readings
	1/26/2024	LAB 1: Introduction to population modeling in Excel, InsightMaker, and R	
Week 2	1/29/2024	LECTURE: Intro to Population Ecology; Exponential growth	
	1/31/2024	LECTURE: Malthus and exponential growth	Gotelli Chapter 2
	2/2/2024	LAB 1 (cont'd)	
Week 3	2/5/2024	LECTURE: Density-dependent population growth	Gotelli Chapter 2
	2/7/2024	LECTURE: Density-dependent population growth	
	2/9/2024	LAB 2: Density-dependent populations in InsightMaker; MSY	
Week 4	2/12/2024	LECTURE: Passenger pigeon/Allee Effect	Gotelli Chapter 3
	2/14/2024	LECTURE: Age-structured populations	Gotelli Chapter 3
	2/16/2024	LAB 3: Age-structured populations in Excel and InsightMaker	
Week 5	2/19/2024	President's Day (no class)	
	2/21/2024	LECTURE: Age-structured populations	Gotelli Chapter 3
	2/23/2024	LAB 4: Matrix population models in R and InsightMaker	
Week 6	2/26/2024	LECTURE: Matrix population models	Heppell 1998 (Optional)
	2/28/2024	LECTURE: Matrix population models	
	3/1/2024	Work in final project groups: PVA proposals	
Week 7	3/4/2024	LECTURE: Matrix population models	
	3/6/2024	LECTURE: Stochasticity and uncertainty	Regan 2002
	3/8/2024	LAB 5: Stochasticity and uncertainty	
Week 8	3/11/2024	Review for Midterm #1	
	3/13/2024	MIDTERM #1	
	3/15/2024	PVA projects: group meetings (or make alternate arrangements for a group meeting time)	
Week 9	3/18/2024	LECTURE: Stochasticity and uncertainty	
	3/20/2024	LECTURE: Small population paradigm	Caughley

Week	Dates	Topic	Readings
			1994
	3/22/2024	Work on PVA projects (PVA models due Apr 10)	
Week 10	3/25/2024	Spring Break (no class)	
	3/27/2024	Spring Break (no class)	
	3/29/2024	Spring Break (no class)	
Week 11	4/1/2024	LECTURE: Declining population paradigm	Caughley 1994
	4/3/2024	LECTURE: Metapopulations	Gotelli Chapter 4
	4/5/2024	LAB 6: Metapopulation modeling in InsightMaker	
Week 12	4/8/2024	LECTURE: Metapopulations	Gotelli Chapter 4
	4/10/2024	LECTURE: Source-sink dynamics	Griffin et al
	4/12/2024	PVA projects: group meetings (working model and description)	
Week 13	4/15/2024	Review for Midterm #2	
	4/17/2024	MIDTERM #2	
	4/19/2024	LAB 7 (optional-no assignment): Parameter estimation: mark-recapture data	
Week 14	4/22/2024	LECTURE: Species interactions: competition	Gotelli Chapter 5
	4/24/2024	LECTURE: Species interactions: competition	Gotelli Chapter 5
	4/26/2024	LAB: Final Project Peer Review (submit peer review)	
Week 15	4/29/2024	LECTURE: Species interactions: predator-prey	Gotelli Chapter 6
	5/1/2024	LECTURE: STUDENT PRESENTATIONS	
	5/3/2024	LAB: STUDENT PRESENTATIONS	
Week 16	5/6/2024	LECTURE: Parameter estimation	
	5/8/2024	NO CLASS: Prep Day	
Week 17	5/13/2024	FINAL EXAM (10:15am to 12:15pm)	
	5/15/2024	FINAL PAPERS DUE	

University Policies

Statement on Academic Dishonesty

The University Academic Standards Policy defines academic dishonesty, and mandates specific sanctions for violations. See the University Academic Standards policy: UAM 6,502

Statement of Disability Services

Any student with a disability needing academic adjustments or accommodations is requested to speak with the instructor or the Disability Resource Center (Pennington Achievement Center Suite 230) as soon as possible to arrange for appropriate accommodations. This course may leverage 3rd party web/multimedia content, if you experience any issues accessing this content, please notify your instructor.

Statement on Audio and Video Recording

Surreptitious or covert video-taping of class or unauthorized audio recording of class is prohibited by law and by Board of Regents policy. This class may be videotaped or audio recorded only with the written permission of the instructor. In order to accommodate students with disabilities, some students may have been given permission to record class lectures and discussions. Therefore, students should understand that their comments during class may be recorded.

Class sessions may be audio-visually recorded for students in the class to review and for enrolled students who are unable to attend live to view. Students who participate with their camera on or who use a profile image are consenting to have their video or image recorded. If you do not consent to have your profile or video image recorded, keep your camera off and do not use a profile image. Students who un-mute during class and participate orally are consenting to have their voices recorded. If you do not consent to have your voice recorded during class, keep your mute button activated and only communicate by using the “chat” feature, which allows you to type questions and comments live.

This is a safe space

The University of Nevada, Reno is committed to providing a safe learning and work environment for all. If you believe you have experienced discrimination, sexual harassment, sexual assault, domestic/dating violence, or stalking, whether on or off campus, or need information related to immigration concerns, please contact the University’s Equal Opportunity & Title IX office at 775-784-1547. Resources and interim measures are available to assist you. For more information, please visit the Equal Opportunity and Title IX page.

Statement on Campus Closures or Delays

In the event of class cancellations or delays caused by inclement weather conditions, fire/smoke conditions, or other unforeseen emergencies, the safety and well-being of students are the University's top priority. Official notifications will be disseminated through the University website and other official channels with details related to any campus delays or closures. In the event of a campus closure, you will be informed as to whether the class will be offered remotely or if it will be canceled. If the class is cancelled, you will receive information on how to address any missed course content. Students facing significant impacts due to these events are encouraged to communicate with their instructor for potential accommodations.

Statement on Failure to Comply with Policy (including as outlined in this Syllabus) or Directives of a University Employee:

In accordance with section 6,502 of the University Administrative Manual, a student may receive academic and disciplinary sanctions for failure to comply with policy, including this syllabus, for failure to comply with the directions of a University Official, for disruptive behavior in the classroom, or any other prohibited action. "Disruptive behavior" is defined in part as behavior, including but not limited to failure to follow course, laboratory or safety rules, or endangering the health of others. A student may be dropped from class at any time for misconduct or disruptive behavior in the classroom upon recommendation of the instructor and with approval of the college dean. A student may also receive disciplinary sanctions through the Office of Student Conduct for misconduct or disruptive behavior, including endangering the health of others, in the classroom. The student shall not receive a refund for course fees or tuition.